Beyond Sparsity: Receptive Field Expansion and Cross-Task Fusion for LiDAR Multi-Task Perception

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***Abstract*—LiDAR-based multi-task perception for autonomous driving requires efficient integration of spatial context and cross-task information, yet existing methods often suffer from restricted receptive fields and suboptimal task interaction. This paper presents a novel multi-task framework that goes beyond sparsity constraints, leveraging receptive field expansion and cross-task fusion to enhance 3D object detection and semantic segmentation. We introduce the Spatial Density-Invariant Multi-scale Integrator (SDIMI), which adaptively fuses multi-resolution contextual features using density-agnostic strategies to expand the receptive field while preserving feature sparsity. Additionally, the Synergistic Instance-Driven Multitask Fusion (SIDMF) module dynamically aligns instance-level features between segmentation and detection, enabling efficient high-level feature propagation via bounding box mapping masks. Experiments on NuScenes and Waymo Open Dataset demonstrate state-of-the-art performance: our method achieves 72.0% NuScenes Detection Score (NDS) and 84.4% mIOU on NuScenes, and 79.8% mAPH-L2 and 72.3% mIOU on Waymo.**

***Index Terms*—Multi-task learning, cross-task fusion, semantic segmentation, object detection.**

I. Introduction

AUTONOMOUS driving significantly depends on accurate environmental perception to ensure safe and efficient navigation [1]. LiDAR's high-precision 3D capabilities make it a critical tool to perform perception tasks like object detection and semantic segmentation. For object detection [2], the task entails identifying and localizing objects using 3D bounding boxes. By comparison, semantic segmentation maps semantic labels to each point in 3D space [3]. With the availability of large-scale datasets, researchers have developed and optimized specialized models for the aforementioned tasks. However, implementing all these single-task solutions is computationally intensive and lacks the ability to generalize across tasks, making their deployment on hardware with limited resources impractical.

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Multi-task learning (MTL) has emerged as a promising paradigm with substantial benefits [4]. By training a single model to perform multiple tasks simultaneously, MTL can leverage shared representations of the input data, thereby reducing computational demands and enhancing generalization across tasks. Existing multi-task perception models typically transform unordered point cloud data into structured representations, such as voxel grids, bird's-eye views (BEV), or range views (RV), which facilitate more efficient feature extraction and learning processes [5]. With the advent of sparse convolution techniques [6], voxel-based methods [7], [8], [9] have become increasingly dominant in the field of 3D perception. These sparse convolution methods compute activations only for non-empty regions of the voxel grid, providing notable computational efficiency advantages. Despite the significant advancements in multi-task perception frameworks, several challenges remain to be addressed in this domain.

A significant limitation arises from the restricted receptive field, which is a consequence of submanifold convolution [6]. Submanifold convolution is applied only to regions where the convolution center has a non-zero response, while ensuring that the input and output feature maps maintain consistent dimensions. However, this selective activation leads to a constrained effective receptive field, thereby restricting the network's ability to capture long-range dependencies and spatial information. VoxelNeXt [10] presents this weakness and enhances the sparse backbone using additional down-sampling layers. However, this operation will alter the density of input sparse features, thereby compromising the inverse sparse convolution in the decoder stage. While LiSD [17] preserves the sparsity of the segmentation branch via trilinear interpolation, the sparsity issue in the detection branch remains unresolved. LidarMultiNet [11] and LiDARFormer [5] transform 3D sparse features into 2D BEV dense features to extract global information. The former employs the Global Context Pooling module for this purpose, while the latter utilizes the cross-space transformer module. However, due to the interference from inactive points, directly extracting global information from the dense BEV view will increase memory consumption. Efficiently expanding the model's receptive field and extracting global information remains an unsolved challenge.

Furthermore, many existing methods [2], [4], [12] primarily focus on using one or more perception tasks as auxiliary tasks, often failing to fully exploit the potential for information exchange between tasks. For instance, PolarStream [2]

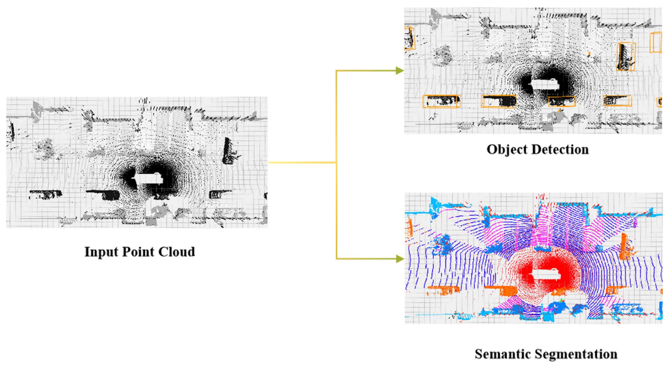

Fig. 1. This paper proposes an efficient LIDAR-based multi-task perception network capable of simultaneously performing semantic segmentation and object detection.

demonstrates the feasibility of integrating 3D object detection, LiDAR segmentation, and panoramic segmentation within a unified framework. However, this approach utilizes task-specific branches for each individual task, with only a minimal shared pillar encoder, thereby constraining the degree of cross-task information sharing. Similarly, both LidarMTL [12] and AOP-Net [4] enable information sharing only among shallow features, which limits the depth of inter-task interaction. Various methods [5], [13], [14] use the cross-task self-attention layer to facilitate information exchange between tasks. While these approach alleviate the computational overhead typically associated with transformer architectures, they inadvertently introduce redundant information and noise due to interactions with irrelevant regions. Additionally, the correlation between queries and tokens remains relatively weak, limiting the effectiveness of cross-task interactions.

To address the aforementioned issues, as shown in Fig. 1 this paper proposes an efficient LIDAR-based multi-task perception network capable of simultaneously performing semantic segmentation and object detection. We introduce the Spatial Density-Invariant Multi-scale Integrator (SDIMI) module, which employs density-agnostic fusion strategies to hierarchically integrate multi-resolution contextual features. This approach effectively increases the network's receptive field while maintaining the invariance of the feature space density. Additionally, we propose the Synergistic Instance-Driven Multitask Fusion (SIDMF) module, which leverages segmentation-to-detection bounding box mapping masks to guide cross-task feature propagation. This module dynamically aligns instance-level features across semantic segmentation and object detection, enabling efficient fusion of high-level features in a multi-task network.

To rigorously validate our model's effectiveness, we conducted comprehensive experiments on two authoritative autonomous driving benchmarks. The proposed method establishes state-of-the-art performance across both perception tasks. Specifically, on the NuScenes dataset [15], it achieves 72.0% NuScenes Detection Score (NDS) for object detection and 84.4% mean Intersection-over-Union (mIOU) for semantic segmentation. On the Waymo Open Dataset [16], our framework demonstrates 79.8% L2 mean Average Precision weighted by Heading (mAPH-L2) for detection and 72.3% mIOU for segmentation, showing consistent superiority across different sensor configurations.

The contributions of this work are summarized as follows:

- We propose an efficient LiDAR multi-task perception network that unifies semantic segmentation and object detection.
- We propose a Spatial Density-Invariant Multi-scale Integrator (SDIMI), which enables the adaptive fusion of multi-scale contextual features by generating spatial attention weights while preserving feature density. This design facilitates the effective deployment of submanifold sparse convolution and expands the network's receptive field.
- We introduce a Synergistic Instance-Driven Multitask Fusion (SIDMF) module that employs instance-level estimation to recover object-specific features and selectively propagates detection-enhanced features to boost semantic segmentation accuracy.
- Extensive experiments demonstrate our framework's superiority, achieving state-of-the-art results on two widely adopted large-scale LiDAR benchmarks.

II. Related Work

A. 3D Object Detection

Lidar point cloud perception involves transforming large-scale sparse point clouds into various representations, such as voxel grids [17], bird's-eye views (BEV) [18], or range views (RV) [19]. Considering the need to balance accuracy with computational efficiency, this paper focuses on the development of voxel-based feature representation approaches. VoxelNet [7] pioneers the discretization of point clouds into 3D grids with 3D convolutional operations, laying the foundation for subsequent voxel-based methodologies. Building on this, SECOND [8] enhances efficiency and accuracy by introducing sparse voxel convolution and an optimized Region of Interest (ROI) feature extraction strategy. PointPillars [9] improves inference speed by discretizing the height dimension into a single layer and utilizing 2D convolutional neural networks for processing pseudo-images. CenterPoint [20] introduces an anchor-free detection paradigm, achieving object detection through heatmap classification, thus constraining the parameter search space. Recent advancements in attention mechanisms have garnered growing interest [21], [22]. SWFormer [23] enables adaptive feature querying within windows via 3D point cloud-to-sparse voxel mapping and window transformation. Flatformer [24] explores self-attention mechanisms under conditions of identical neighborhood window dimensions but varying spatial configurations. In FocalFormer3D [25], the attention mechanism discovers these hard-to-detect objects through multi-stage query generation and effectively distinguishes numerous candidate proposals via a box-level transformer decoder.

B. 3D Semantic Segmentation

Similar to object detection, semantic segmentation of point clouds includes point-based [26], [27], range-view based [28], [29], and voxel-based methods [30], [31], and we mainly focus on introducing voxel-based methods. SegCloud [31] uses trilinear interpolation to integrate voxel features with Conditional Random Fields(CRF) for learning fine-grained 3D data predictions. PointGrid [32], by employing improved sampling strategies, achieves a structured and consistent distribution of point clouds in each voxel. This method effectively addresses the limitations of conventional methods due to resolution constraints. Cylinder3D [33] leverages cylindrical partitioning to extract features at the voxel level, employing asymmetrical residual blocks and the dimension-decomposition based context modeling module to address the challenges posed by the sparsity and uneven distribution of point clouds. To mitigate the computational cost of 3D convolution, Graham [6] introduces an efficient convolution technique called Submanifold Sparse Convolution. This method performs convolution only at positions where a non-zero activation is present at the convolution center, thereby avoiding unnecessary computations in empty regions and preserving the sparsity of voxel features. Recent studies [34], [35] along this line confirm the effectiveness of this operation. However, selective convolution may impose limitations on both the receptive field of the model and the integration of global information.

C. Multi-Task LiDAR Perception

Numerous works [2], [4], [12], [36] have attempted to employ the hard parameter sharing scheme [37] to simultaneously perform object detection and semantic segmentation. These networks typically consist of an encoder-decoder structure, where the shared encoder extracts feature information and decoder branches generate task-specific outputs. DWSNet [38] implicitly integrates semantic information via segmentation loss to achieve improved detection accuracy. LidarMTL [12] proposes a multi-task architecture incorporating object detection and multiple road perception tasks. However, these works primarily employ auxiliary tasks to provide inductive bias. By refining the shallow layers, the model tends to explain the assumptions commonly across multiple tasks. Currently, there are ongoing efforts to explore further cross-task information interaction. AOP-Net [4] integrates a 3D object detection model based on BEV with 3D panoramic segmentation model based on RV, and it preliminarily explores an instance-based feature retrieval (IFR) module. This module enables feature recovery using predicted instance masks, thereby enhancing the detection backbone. Similarly, LISD [17] uses 3D sparse convolution as a feature extractor. An instance-aware refinement module is used for cross-task information interaction, which improves the feature representation by the object suggestion. LidarFormer [5] significantly improves detection accuracy by transferring specific task prediction features within the task decoder through a query-based approach.

III. Method

This section presents a comprehensive introduction of the proposed multi-task network. It is organized into four parts: the overall architecture, the implementation of the Spatial Density-Invariant Multi-scale Integrator (SDIMI (Section III-B)), the design of the Synergistic Instance-Driven Multitask Fusion (SIDMF (Section III-C)), the loss function used during network training (Section III-D).

A. Overall Architecture

Following the widely adopted paradigm in [5] [11] [17], our model employs a unified voxel-based sparse convolutional framework with task-specific branches as shown in Fig. 2. It receives LiDAR-based scene information, represented as an unordered point cloud $\mathcal{P} = \{p_i \mid p_i \in \mathbb{R}^{3+c}\}_{i=1}^{N}$, where N denotes the number of points. Each point $p_i \in \mathbb{R}^{3+c}$ is characterized by 3D coordinates (x, y, z) and c additional features, such as reflection intensity, timestamp, LiDAR elongation. The segmentation branch predicts the label of each point $L = \{l_i \mid l_i \in (1 \dots K)\}_{i=1}^{N}$, where K represents the number of segmentation classes. The detection branch predicts the 3D oriented bounding box $O = \{o_i \mid o_i \in \mathbb{R}^7\}_{i=1}^{B}$ and the class label $C = \{c_i \mid c_i \in (1 \dots M)\}_{i=1}^{B}$ for all objects in the scene, where B, M represent the number of object and detection classes.

Voxelization transforms unordered 3D point cloud data into a 3D grid while preserving key information. This is done by transforming the real-valued coordinates (x, y, z) of the point p_i into discrete voxel grid coordinates. The voxel coordinates (i, j, k) are given by:

$$i = \left\lfloor \frac{x - x_{\min}}{\Delta x} \right\rfloor, \quad j = \left\lfloor \frac{y - y_{\min}}{\Delta y} \right\rfloor, \quad k = \left\lfloor \frac{z - z_{\min}}{\Delta z} \right\rfloor \tag{1}$$

where $(x_{\min}, y_{\min}, z_{\min})$ is the minimum bounds of the point cloud. $\lfloor . \rfloor$ denotes the floor function that rounds down to the nearest integer. $(\Delta x, \Delta y, \Delta z)$ define the voxel resolution.

Occupied voxels are fed into a backbone with an encoder-decoder structure for further extraction of LiDAR features. In the encoder, we utilize a modified version of VoxelNeXt [10], which incorporates four groups of convolutional blocks, each consisting of 3D sparse convolutions and submanifold convolutions. This reduces the input spatial resolution to 1/8, simultaneously extracting high-level features. Unlike VoxelNeXt, which increases the receptive field of the encoder and extracts global information through additional down-sampling layers, we achieve the same effect using the SDIMI(Section III-B) module. Additionally, this approach not only maintains feature sparsity but also enables the adaptive fusion of multi-scale features. In the decoder, 3D sparse deconvolutions progressively upsample the LiDAR features to the original spatial resolution and fuse them with skip-connections from the encoder to preserve fine-grained details.

To effectively leverage the features from the 3D backbone for object detection, we follow the strategy of LidarMulti-Net [11] by collapsing the 3D features from the final encoder stage along the height dimension. These features are mapped to BEV and processed by a 2D backbone network to capture rich contextual information. To further enhance the model's capability in capturing global dependencies and fine-grained details, we employ a Transformer-based detection head adapted from BEVFusion [39]. Specifically, we adopt a query-based mechanism rather than processing the entire dense feature map. We first identify the top-T object candidates by selecting the highest-scoring local maxima from the initial heatmap predictions. Feature vectors at these candidate locations are sampled from the BEV feature map to initialize the object queries. To

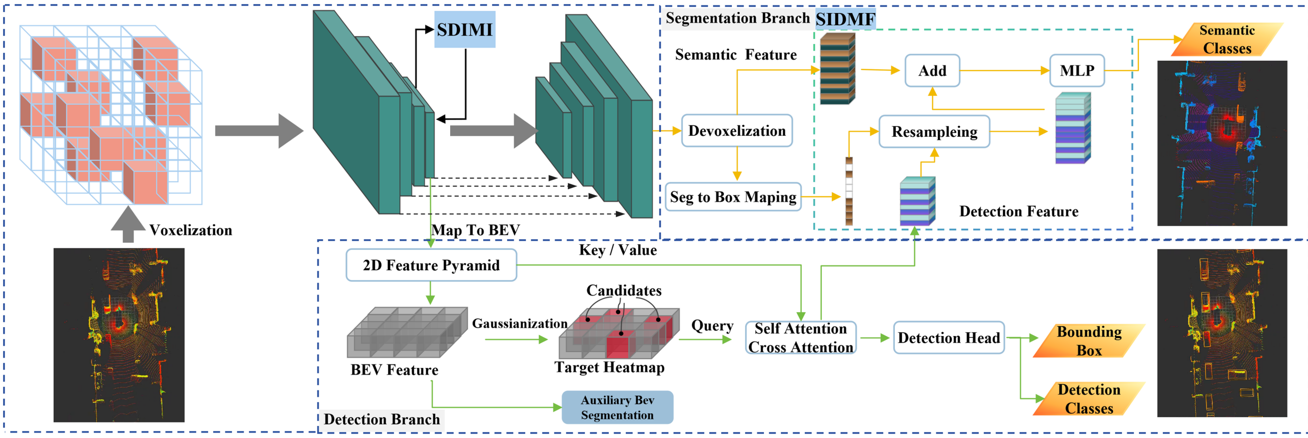

Fig. 2. Architecture of the proposed LiDAR multi-task perception algorithm: The workflow involves voxelization of LiDAR point clouds, feature extraction via an encoder-decoder network, expansion of the receptive field using the Spatial Density-Invariant Multi-scale Integrator (SDIMI), cross-task feature fusion through the Synergistic Instance-Driven Multitask Fusion (SIDMF) module, and parallel branches for 3D object detection (with bird's-eye view (BEV) mapping) and semantic segmentation.

encode semantic distinctions, learnable category embeddings are added element-wise to these queries. Subsequently, the queries are processed by a Transformer decoder layer comprising two key components: (1) a *self-attention module* that models inter-query interactions to suppress duplicate detections, and (2) a *cross-attention module* that aggregates global context. In the cross-attention mechanism, object queries serve as the *Query*, while the flattened global BEV features act as both the *Key* and *Value*. Learned positional embeddings are incorporated into both the queries and BEV features to preserve spatial structure. Finally, the refined queries are passed through Feed-Forward Networks to regress bounding box attributes and classification scores. Additionally, we incorporate an auxiliary BEV segmentation head to provide coarse semantic guidance. For training supervision, the ground truth class of each BEV cell is determined via majority voting of the annotated LiDAR points within that cell.

For the segmentation branch, following the design of PVConv [40], we employ trilinear interpolation to devoxelize voxel-wise features obtained from the decoder into point-wise features, based on the relative distance between each point and its surrounding voxel grid. Afterward, the SIDMF (Section III-C) module is applied to enhance the segmentation features at the object level, and the per-point segmentation categories are subsequently predicted.

B. Spatial Density-Invariant Multi-Scale Integrator

According to VoxelNext [10], the receptive field size plays a crucial role in determining whether a model based on sparse voxel features can effectively capture long-range contextual information, thereby enhancing prediction performance. To enlarge the receptive field, VoxelNext introduces two additional downsampling layers. The output feature F_4 from the final encoder has a resolution of 1/8 relative to the original input, while the two additional layers produce features F_5 and F_6 with resolutions of 1/16 and 1/32, respectively. Drawing upon the common practices described in [10] and [17], the features

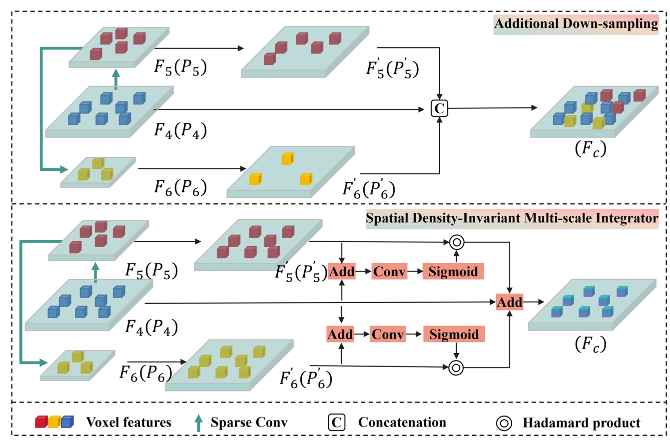

Fig. 3. Comparative architecture of additional downsampling (top) and the proposed Spatial Density-Invariant Multi-scale Integrator (SDIMI) (bottom): The top design shows the consecutive downsampling of VoxelNeXt and LiSD's detection branch for receptive field expansion, which incurs feature density variation; the bottom design presents SDIMI's density-agnostic and adaptive multi-scale integration framework.

$\{F_4, F_5, F_6\}$ are subsequently fused to F_c following the Additional Down-sampling part illustrated in Fig. 3, which is formulated as:

$$
\begin{aligned}
F_c &= F_4 \cup (F_5 \cup F_6)\,, \\
P_6' &= \left\{ \left(x \times 2^2, y \times 2^2, z \times 2^2\right) \mid \mathbf{x} \in P_6 \right\} \\
P_5' &= \left\{ \left(x \times 2^1, y \times 2^1, z \times 2^1\right) \mid \mathbf{x} \in P_5 \right\} \\
P_c &= P_4 \cup (P_5' \cup P_6')\,,
\end{aligned} \tag{2}
$$

where P_i denotes the set of spatial locations associated with the features F_i, each point $\mathbf{x} \in P_i$ is represented by 3D coordinates $\{x, y, z\}$. $\mathbf{x}' \in P_i'$ denotes the set of aligned locations. The features from P_5 and P_6 are aligned to P_4 based on their respective downsampling ratios. Although this minor modification significantly enlarges the receptive field, it also causes a noticeable change in feature density, which is unfavorable for the further application of sparse convolution in the decoder. In addition, simply concatenating multi-level features fails to adequately integrate multi-scale information.

To address the aforementioned issues, we propose an improvement to VoxelNeXt [10] and LiSD [17] that effectively enlarges the receptive field while preserving dual-branch sparsity and enabling adaptive multi-scale feature fusion. As illustrated in the lower part of Fig. 3, we similarly downsample P_4 twice to generate P_5 and P_6. During the alignment of P_5 and P_6 to the resolution of P_4, trilinear interpolation is applied:

$$F_i'(\mathbf{x}') = \sum_{i=0}^{1}\sum_{j=0}^{1}\sum_{k=0}^{1} w_{ijk} \cdot F_i\left(\lfloor x' \rfloor + i, \lfloor y' \rfloor + j, \lfloor z' \rfloor + k\right), \tag{3}$$

where F_i' denotes the set of aligned feature. Feature sparsity is preserved as interpolation is only applied to the non-empty voxels in F_4. The weights w_{ijk} are computed as:

$$\begin{aligned} w_{ijk} = &\ (1 - |x' - (\lfloor x' \rfloor + i)|) \\ &\cdot (1 - |y' - (\lfloor y' \rfloor + j)|) \cdot (1 - |z' - (\lfloor z' \rfloor + k)|). \end{aligned} \tag{4}$$

Spatial attention weights enable the adaptive selection of salient regions across multi-scale features. To compute these attention weights, we first sum the feature maps F_4 and F_i' element-wise, and then apply a convolution operation, as defined below:

$$A_i = \sigma\left(f^{5\times 5}\left(F_4 + F_i'\right)\right), \tag{5}$$

where σ denotes the Sigmoid function, and $f^{5\times 5}$ refers to a convolution with a 5×5 kernel. The spatial attention weights encode the importance of regions to be either emphasized or suppressed. The final fused feature is computed as :

$$F_c = f^{5\times 5}\left(F_4 + A_5 \odot F_5' + A_6 \odot F_6'\right) \tag{6}$$

where $\odot$ denotes Hadamard product.

C. Synergistic Instance-Driven Multitask Fusion

While semantic segmentation excels at assigning fine-grained class labels to every point in LiDAR point cloud, it often struggles to distinguish between individual object instances in cluttered environments. In contrast, object detection, particularly with anchor-free or Gaussian-based instance estimation, focuses on identifying distinct object-level features and spatial priors. As illustrated in the SIDMF part in Fig. 2, our insight is to utilize these high-quality instance representations from the detection branch to guide and refine the segmentation features. This mutual collaboration addresses the inherent limitations of purely per-point segmentation and enriches the semantic understanding of the scene.

Let $F_{\text{det}} \in \mathbb{R}^{T\times C_d}$ represent the instance-level features for the top-T objects predicted by the detection branch, where C_d is the channel dimension. These shallow features encapsulate rich spatial and semantic information about distinct object candidates, usually centered at the predicted Gaussian heatmap peaks. The segmentation branch outputs high-resolution features for each point p_i, reconstructed through a de-voxelization layer from voxel features $V \in \mathbb{R}^{H\times W\times D\times C_s}$, where (H, W, D) define the voxel size and C_s is the semantic channel dimension. This produces pointwise features $F_{\text{seg}} \in \mathbb{R}^{N\times C}$.

Inspired by the point-wise masks in LidarMultiNet [11] and LiSD [17], SIDMF maps points to detected objects for selective instance-level fusion. Unlike these methods, we preserve the background to suppress detection noise. Specifically, for each point p_i, we determine whether it lies inside one or more predicted bounding boxes $O_i \in \mathbb{R}^7$. This yields a binary indicator matrix $M \in \{0,1\}^{N\times T}$, where:

$$M_{ij} = \begin{cases} 1, & \text{if } p_i \in O_j \\ 0, & \text{otherwise} \end{cases}. \tag{7}$$

We compute enhanced segmentation features by integrating instance-level features with point features $F_{\text{seg}}^{\text{enh}} \in \mathbb{R}^{N\times C_s}$ by integrating instance-level features with point features. For each point p_i if it falls into multiple boxes, the associated instance features are averaged. Formally:

$$F_{\text{seg}}^{\text{enh}}(p_i) = F_{\text{seg}}(p_i) + \frac{1}{\sum_{j=1}^{T} M_{ij}} \sum_{j=1}^{T} M_{ij} \cdot \phi\left(F_{\text{det}}(j)\right), \tag{8}$$

where $\phi : \mathbb{R}^{C_d} \to \mathbb{R}^{C_s}$ is a learnable MLP designed to align dimensions and filter cross-task features. The shallow features and averaging operation are introduced to smooth out noise interference across different detection features. Compared to concatenation, addition avoids the computational burden of dimension expansion. This residual addition encourages the segmentation head to attend more to object boundaries, shapes, and priors learned by the detection branch.

Finally, the enhanced segmentation features $F_{\text{seg}}^{\text{enh}} \in \mathbb{R}^{N\times C_s}$ are passed through an MLP classifier to predict per-point semantic classes $\hat{l}_i$:

$$\hat{l}_i = \text{MLP}\left(F_{\text{seg}}^{\text{enh}}(p_i)\right), \quad \hat{l}_i \in (1 \ldots K). \tag{9}$$

D. Loss Function

Our network is trained end-to-end using a multi-task loss function that jointly optimizes the objectives of 3D object detection, semantic segmentation, and auxiliary BEV segmentation.

In the detection branch, following LidarMultiNet [11], heatmap prediction is trained with a focal loss $\mathcal{L}_{\text{hm}}$ [41] to handle the foreground-background imbalance. Size, orientation, and IoU refinement are trained with L1 regression losses, denoted as $\mathcal{L}_{\text{reg}}$ and $\mathcal{L}_{\text{iou}}$, respectively. The total detection loss is thus:

$$\mathcal{L}_{\text{det}} = \mathcal{L}_{\text{hm}} + \mathcal{L}_{\text{reg}} + \mathcal{L}_{\text{iou}} \tag{10}$$

For semantic segmentation and auxiliary BEV segmentation, we apply standard cross-entropy loss over the predicted point-wise and BEV semantic classes, respectively:

$$\mathcal{L}_{\text{seg}} = \text{CE}\left(l_{\text{seg}}, \hat{l}_{\text{seg}}\right), \quad \mathcal{L}_{\text{bev}} = \text{CE}\left(l_{\text{bev}}, \hat{l}_{\text{bev}}\right) \tag{11}$$

where CE(.) denotes the cross-entropy function between ground-truth and predicted labels.

The overall training objective is formulated as a weighted sum of the task-specific losses:

$$\mathcal{L}_{\mathrm{MTL}} = w_1 \cdot \mathcal{L}_{\mathrm{det}} + w_2 \cdot \mathcal{L}_{\mathrm{seg}} + w_3 \cdot \mathcal{L}_{\mathrm{bev}} \tag{12}$$

where w_1, w_2, w_3 are weights that control the influence of each task. We adopt the task uncertainty weighting strategy proposed by Kendall et al. [42], which dynamically balances the task losses by modeling homoscedastic uncertainty. This approach introduces a learnable noise parameter σ for each task, resulting in the following adaptive multi-task loss:

$$\mathcal{L}_{\mathrm{MTL}}^{\mathrm{adaptive}} = \sum_{i \in \{\mathrm{det},\mathrm{seg},\mathrm{bev}\}} \frac{1}{2\sigma_i^2}\mathcal{L}_i + \frac{1}{2}\log \sigma_i^2. \tag{13}$$

This adaptive formulation allows the network to learn the relative weighting of each task based on its predictive uncertainty, improving stability and convergence in multi-task learning settings.

IV. Experiments

A. Dataset

We evaluate our proposed multi-task perception framework on two large-scale autonomous driving datasets: nuScenes [15] and the Waymo Open Dataset (WOD) [16]. Both datasets provide high-quality 3D LiDAR point clouds with object-level and point-level annotations, making them well-suited for joint 3D object detection and semantic segmentation tasks.

The nuScenes dataset comprises 1,000 driving scenes, each lasting 20 seconds. Following the official split, we use 700 scenes for training, 150 for validation, and 150 for testing. Detection performance is evaluated using mean Average Precision (mAP) and nuScenes Detection Score (NDS), while mean Intersection-over-Union (mIoU) is used for segmentation.

The Waymo Open Dataset consists of 1,150 sequences, with 798 for training, 202 for validation, and 150 for testing. For detection, we follow the official Waymo evaluation protocol using Average Precision weighted by heading (APH). Ground-truth objects are stratified into LEVEL 1 (L1) and LEVEL 2 (L2) categories based on factors such as point density and occlusion. Specifically, APH-L1 includes only L1 samples, while APH-L2 incorporates both L1 and L2 samples. Segmentation performance is reported using mIoU across 23 annotated classes.

B. Experiment Setup

We implement our framework using PyTorch [43] and the OpenPCDet [44] toolbox. To ensure a consistent and rigorous comparative analysis, many of our experimental hyperparameters are configured in reference to the settings used in LiSD [17]. The network is trained from scratch for 70 epochs using the AdamW optimizer with a weight decay of 10^{-2}. The learning rate follows the One Cycle policy, starting at 10^{-3}. Training is conducted on 4 NVIDIA A30 GPUs with a batch size of 8.

Voxelization and Temporal Fusion: We adopt dataset-specific spatial configurations for voxelization. For the nuScenes dataset, the point cloud range is set to $[-54.0, 54.0]$ m along the x and y axes, and $[-5.0, 4.6]$ m along the z axis, with a voxel resolution of $[0.075, 0.075, 0.2]$ m. To leverage temporal information, we concatenate the current frame with the previous 9 LiDAR sweeps. For the Waymo Open Dataset, we use a point cloud range of $[-75.2, 75.2]$ m in the horizontal plane and $[-2.0, 4.6]$ m along the vertical axis, with a voxel size of $[0.1, 0.1, 0.15]$ m. The current frame is fused with the preceding 2 LiDAR sweeps to capture short-term motion cues.

Data Augmentation: To improve generalization, we apply a combination of geometric augmentations during training. These include random flipping along the xz and yz planes, global scaling with a factor uniformly sampled from $[0.95, 1.05]$, and global rotation within the range $[-\frac{\pi}{4}, \frac{\pi}{4}]$. In addition, we apply random translation along all three axes, where the displacement Δd is drawn from a normal distribution $\mathcal{N}(0, 0.5^2)$.

C. Object Detection Result

To validate the effectiveness of our proposed multi-task LiDAR perception framework, we conduct extensive evaluations on two large-scale benchmarks: nuScenes and Waymo. As shown in Table I, our method significantly outperforms several state-of-the-art 3D object detectors, including single-task models such as PV-RCNN [45], CenterPoint [20], TransFusion-L [14], VoxelNeXt [10], FocalFormer3D [25], and DSVT [48], as well as representative multi-task frameworks like LidarFormer [5] and LidarMultiNet [11].

Our method achieves the highest detection performance on the nuScenes dataset. On the validation set, it reaches 67.1% mAP and 72.0% NDS, surpassing the previous best single-task method (DSVT) by 0.7% and 0.9%, respectively. More importantly, when compared against specialized multi-task frameworks on nuScenes, our model demonstrates a clear advantage. It outperforms LidarFormer (66.6% mAP, 70.8% NDS) and LidarMultiNet (63.8% mAP, 69.5% NDS) by a notable margin. This superiority is maintained on the test set, where we achieve 69.9% mAP and 73.5% NDS, surpassing LidarFormer (68.9% mAP, 72.4% NDS) and setting a new state of the art for accuracy.

The superiority of our approach is further reinforced on the Waymo Open Dataset. Our method delivers leading APH across all categories at both L1 and L2 difficulty levels. Notably, our model achieves 79.8% L2 mAPH, significantly outperforming multi-task baselines such as LidarFormer (76.2%) and LidarMultiNet (75.2%) by 3.6% and 4.6%, respectively. These results highlight our model's robust capability to capture complex spatial features across different sensor configurations and datasets.

In terms of inference speed, our method maintains a competitive latency of 107 ms on nuScenes. While some single-task models like CenterPoint and VoxelNeXt exhibit lower latencies, they lack the capability to perform segmentation. Critically, our framework is significantly more efficient than other multi-task models; it operates faster than both LidarFormer (168 ms) and LidarMultiNet (163 ms) while providing higher precision. This demonstrates that our architecture offers a much more powerful, versatile, and computationally efficient solution for autonomous driving by successfully unifying detection and segmentation tasks.

TABLE I

QUANTITATIVE COMPARISON OF 3D DETECTION PERFORMANCE ON NUSCENES AND WAYMO. LATENCIES ARE MEASURED ON AN NVIDIA A30 GPU. BOLD AND UNDERLINED INDICATE THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY. NO TEST-TIME AUGMENTATION (TTA) IS APPLIED.

Method	nuScenes [val]			nuScenes [test]		Waymo [val]							
	mAP	NDS	Latency (ms)	mAP	NDS	Veh (APH) L1	Veh (APH) L2	Ped (APH) L1	Ped (APH) L2	Cyc (APH) L1	Cyc (APH) L2	mAPH L1	mAPH L2
PV-RCNN [45]	-	-	150	-	-	76.9	68.4	65.6	57.6	66.4	64.0	69.6	63.3
CenterPoint [20]	57.1	65.4	73	58.0	65.5	70.8	62.7	65.5	58.2	67.4	64.9	67.9	61.9
TransFusion-L [14]	64.7	69.3	108	65.5	70.2	-	65.1	-	64.0	-	67.4	-	65.5
VoxelNeXt [10]	60.5	66.6	**67**	66.2	71.4	77.7	69.4	76.3	68.6	74.9	72.2	76.3	70.1
LargeKernel3D [46]	63.3	69.1	157	65.3	70.6	-	-	-	-	-	-	77.6	69.4
FocalFormer3D [25]	66.4	70.9	129	68.7	72.6	-	67.6	-	66.8	-	72.6	-	69.0
Point Transformer V3 [47]	-	-	-	-	-	-	72.5	-	75.0	-	71.4	-	73.0
LidarFormer [5]	66.6	70.8	168	68.9	72.4	-	-	-	-	-	-	-	76.2
LidarMultiNet [11]	63.8	69.5	163	67.0	71.6	-	-	-	-	-	-	-	75.2
DSVT [48]	66.4	71.1	84	68.4	72.7	81.6	74.1	83.2	76.4	80.3	78.0	81.7	76.2
Ours	**67.1**	**72.0**	107	**69.9**	**73.5**	**87.3**	**80.4**	**85.2**	**79.5**	**81.8**	**79.4**	**84.8**	**79.8**

TABLE II

PER-CLASS SEMANTIC SEGMENTATION RESULTS ON NUSCENES VAL SET. THE BEST AND SECOND-BEST RESULTS ARE MARKED IN **BOLD** AND UNDERLINED, RESPECTIVELY.

Method	*mIoU*	*Latency*	*barrier*	*bicycle*	*bus*	*car*	*construction*	*motorcycle*	*pedestrian*	*traffic cone*	*trailer*	*truck*	*driveable*	*other flat*	*sidewalk*	*terrain*	*manmade*	*vegetation*
PolarNet [49]	71.0	77	74.7	28.2	85.3	90.9	35.1	77.5	71.3	58.8	57.4	76.1	96.5	71.1	74.7	74.0	87.3	85.4
Cylinder3D [33]	76.1	117	76.4	40.3	91.2	93.8	31.3	78.0	78.9	64.9	62.1	84.4	96.8	71.6	76.4	74.5	90.5	87.4
PVKD [50]	76.0	**73**	76.2	40.0	90.2	**94.0**	50.9	77.4	78.8	64.7	62.0	84.1	96.6	71.4	76.4	73.6	90.3	86.9
RPVNet [1]	77.6	150	78.2	43.4	92.7	92.7	51.5	87.0	81.5	66.0	63.0	84.0	96.9	**73.5**	75.9	76.0	90.6	88.9
SphereFormer [3]	78.4	150	77.7	43.8	**94.5**	93.1	52.4	86.9	81.2	65.4	73.4	**85.3**	97.0	73.4	75.4	75.0	91.0	89.2
UDeerPep [51]	81.8	-	**85.5**	55.5	90.5	91.6	72.2	85.6	81.4	76.3	87.3	74.0	97.7	70.2	72.1	77.4	92.7	90.2
LidarFormer [5]	81.5	168	83.9	41.0	86.2	92.6	71.6	**91.3**	84.5	**81.3**	**89.0**	74.5	**97.9**	69.8	**81.4**	77.6	92.2	89.2
LidarMultiNet [11]	81.2	163	80.2	48.0	94.2	89.7	71.3	86.9	85.0	80.0	86.7	74.5	97.7	67.0	80.5	76.2	92.0	89.3
Ours	**84.4**	107	83.2	**68.7**	91.5	92.8	**81.2**	86.4	**86.9**	80.3	87.7	79.4	**97.9**	71.4	81.3	**77.9**	**93.2**	**90.4**

D. Semantic Segmentation Results

Table II reports the per-class semantic segmentation performance on the nuScenes validation set. We compare our method with several state-of-the-art LiDAR-based segmentation approaches, including dedicated single-task models such as PolarNet [49], Cylinder3D [33], PVKD [50], RPVNet [1], SphereFormer [3], and UDeerPep [51], as well as representative multi-task frameworks including LiDARFormer [5] and LidarMultiNet [11]. Our approach achieves a new state-of-the-art mean IoU (mIoU) of 84.4%, outperforming the previous best single-task method (UDeerPep) by 2.6% and surpassing contemporary multi-task baselines by a significant margin.

In terms of per-class performance, our method obtains the best results in 7 out of 16 classes and maintains highly competitive accuracy across the remaining categories. Specifically, it achieves top scores in bicycle (68.7%), construction (81.2%), pedestrian (86.9%), driveable surface (97.9%), terrain (77.9%), manmade (93.2%), and vegetation (90.4%). Furthermore, our method ranks second-best in traffic cone (80.3%), trailer (87.7%), and sidewalk (81.3%) classes, trailing slightly behind LiDARFormer.

Notably, our framework exhibits a clear advantage over other multi-task perception models. While LiDARFormer and LidarMultiNet achieve 81.5% and 81.2% mIoU respectively, our method outperforms them by approximately 2.9% and 3.2%. These gains are particularly pronounced in challenging categories such as bicycle (+27.7% over LiDARFormer) and construction (+9.6% over LidarMultiNet). This performance gap validates the effectiveness of our SIDMF module, which successfully leverages instance-level priors from the detection branch to refine point-wise semantic reasoning.

Despite its multi-task capability, our framework maintains a competitive latency of 107 ms. It not only outperforms dedicated segmentation models like RPVNet and SphereFormer (150 ms) but is also significantly more efficient than multi-task competitors such as LiDARFormer (168 ms) and LidarMultiNet (163 ms). This highlights the efficiency of our joint-learning architecture, which delivers a more versatile and faster solution for autonomous driving by optimizing shared representations.

E. Comparison With Other Multi-Task Methods

Table III provides a comprehensive comparison with state-of-the-art multi-task methods on the Waymo Open Dataset (WOD) validation split. These methods, including AOP-Net [4], LidarMultiNet [11], LidarFormer [5], and LiSD [17], are designed to simultaneously handle 3D object detection and semantic segmentation.

As shown in Table III, our method achieves a significant breakthrough in detection accuracy, yielding 79.8% mAPH-L2. This result outperforms the previous state-of-the-art method,

TABLE III
COMPARISON WITH OTHER MULTI-TASK METHODS ON THE *VAL* SPLIT OF THE WOD DATASET. THE BEST AND SECOND-BEST RESULTS ARE MARKED IN **BOLD** AND UNDERLINED, RESPECTIVELY. THE SYMBOL † DENOTES THE RESULTS RETRAINED UNDER THE ALIGNED TRAINING SETTINGS WITHOUT TEST-TIME AUGMENTATION AND * INDICATES THE RESULTS OBTAINED BY OUR EXPERIMENTS.

Task	Model	mIoU	mAPH-L2	FLOPs (G)	Latency (ms)*
Multi-task	AOP-Net [4]	66.3	63.7	-	-
	LidarMultiNet [11]	71.9	75.2	183.6	179
	LidarFormer [5]	72.2	<u>76.2</u>	189.4	186
	LiSD [17]	**72.6**	76.1	-	-
	LiSD†	71.9	74.3	<u>174.3</u>	<u>137</u>
	Ours	<u>72.3</u>	**79.8**	**167.3**	**124**

TABLE IV
ABLATION STUDY ON THE NUSCENES VALIDATION SET. WE ANALYZE THE CONTRIBUTION OF EACH PROPOSED COMPONENT TO TASK PERFORMANCE AND COMPUTATIONAL EFFICIENCY. THE LATENCY IS MEASURED ON AN NVIDIA A30 GPU.

seg	det	MTL	SDIMI	$\mathcal{L}_{\text{bev}}$	SIDMF	mIoU	mAP	NDS	Latency (ms)	FLOPs (G)
✓						77.4	-	-	79	98.5
	✓					-	62.8	67.1	69	83.3
✓	✓	✓				79.6	64.4	69.9	92	133.8
✓	✓	✓	✓			80.7	65.1	70.7	98	141.4
✓	✓	✓	✓	✓		82.3	66.5	71.3	-	-
✓	✓	✓	✓	✓	✓	**84.4**	**67.1**	**72.0**	107	156.1

LidarFormer, by a substantial margin of 3.6%. Regarding the semantic segmentation task, our model achieves a highly competitive 72.3% mIoU, which is the second-best result among all compared methods, trailing the top-performing LiSD (72.6%) by only 0.3%.

To further evaluate the architectural efficiency independent of specific hardware and software optimizations, we report the theoretical computational cost in terms of FLOPs. Our framework achieves the lowest computational overhead with 167.3G FLOPs, demonstrating superior efficiency compared to other unified frameworks such as LidarMultiNet (183.6G) and LidarFormer (189.4G). Furthermore, our model operates with a real-time latency of only 124 ms, outperforming the retrained LiSD† (137 ms) and significantly faster than LidarMultiNet (179 ms) and LidarFormer (186 ms). These metrics confirm that our "Beyond Sparsity" is well-suited for high-performance, real-time autonomous driving applications.

F. Ablation Study

To rigorously validate the performance of each proposed component and evaluate their computational efficiency, we conduct comprehensive ablation studies on the validation split of the nuScenes dataset. As detailed in Table IV, we investigate the progressive contribution of individual modules to the overall system accuracy, inference latency, and computational complexity. All efficiency metrics were measured on a single NVIDIA A30 GPU with a batch size of 1.

We first establish the single-task baselines. Both models utilize the backbone defined in Section III-A but exclude our proposed SDIMI module. For segmentation, this model applies devoxelization to the backbone output and uses a simple MLP for classification, without the SIDMF module. For detection, this model projects voxel features to BEV and utilizes the heatmap and self-attention mechanism followed by standard regression heads.

TABLE V
COMPARATIVE ANALYSIS OF DIFFERENT RECEPTIVE FIELD EXPANSION STRATEGIES ON THE NUSCENES VALIDATION SET. "NONE" REPRESENTS THE BASELINE MULTI-TASK MODEL WITHOUT ADDITIONAL DOWNSAMPLING. BEST RESULTS ARE MARKED IN **BOLD**, AND SECOND-BEST RESULTS ARE UNDERLINED.

Method	mIoU	mAP	NDS	FLOPs (G)	Latency (ms)
None	79.6	64.4	69.9	**133.8**	**92**
BEV-DS [11]	79.7	64.8	70.1	142.3	101
Voxel-DS [10]	<u>80.5</u>	<u>65.1</u>	<u>70.4</u>	145.7	105
SDIMI (Ours)	**80.7**	**65.1**	**70.7**	<u>141.4</u>	<u>98</u>

The semantic segmentation model attains 77.4% mIoU with a latency of 79 ms and 98.5G FLOPs. The object detection model achieves 62.8% mAP (67.1% NDS) at 69 ms and 83.3G FLOPs. Simply summing these separate models would result in a prohibitive total latency of 148 ms and 181.8G FLOPs. By integrating the detection loss $\mathcal{L}_{\text{det}}$ and unifying the backbone, the model transitions to a multi-task learning paradigm. This yields a substantial performance boost, elevating mIoU to 79.6%, mAP to 64.4%, and NDS to 69.9%. Crucially, the shared feature extraction significantly improves efficiency, reducing the total latency to 92 ms and the computational cost to 133.8G FLOPs.

Subsequently, we incorporate SDIMI. By adaptively fusing multi-scale contextual features via learned spatial attention, this module effectively expands the receptive field while preserving feature sparsity. This structural optimization results in further improvements, with mIoU rising to 80.7%, mAP to 65.1%, and NDS to 70.7%. In terms of computational cost, SDIMI introduces a marginal increase of 6 ms in latency and 7.6G in FLOPs. This indicates that our density-invariant design efficiently captures global context without imposing the heavy computational burden characteristic of dense operations. To rigorously validate the effectiveness of SDIMI, we conducted a comparative analysis against representative receptive field expansion strategies, as detailed in Table V and visualized in Fig. 4. As shown in

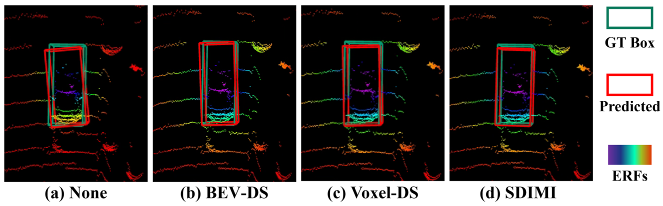

Fig. 4. Visual comparison of different receptive field expansion strategies. (a) The baseline architecture without additional downsampling layers has a limited receptive field. (b) Projecting 3D features to 2D BEV followed by dense downsampling [11]. (c) Standard strided sparse convolution on voxels [10]. (d) Our proposed SDIMI. ERFs: effective receptive fields.

TABLE VI
COMPARISON OF DIFFERENT FEATURE INTERSECTION STRATEGIES ON NUSCENES VALIDATION SET. "NONE" REPRESENTS THE MULTI-TASK MODEL WITHOUT INTERSECTION. BEST RESULTS ARE MARKED IN **BOLD**, AND SECOND-BEST RESULTS ARE <u>UNDERLINED</u>.

Method	mIoU	mAP	NDS	FLOPs (G)	Latency (ms)
None	82.3	66.5	71.3	141.4	98
Addition	82.1	65.8	71.3	144.8	100
Concatenation	82.0	66.1	70.9	146.7	102
GCP [11]	<u>83.5</u>	<u>67.0</u>	<u>71.7</u>	151.6	106
SIDMF (Ours)	**84.4**	**67.1**	**72.0**	156.1	107

the comparison, the BEV-based downsampling strategy (BEV-DS) [11] expands the global receptive field but compromises vertical structural information due to height dimension collapse, yielding only marginal segmentation gains (79.7% mIoU). Alternatively, standard Voxel-based downsampling (Voxel-DS) [10] achieves competitive accuracy (80.5% mIoU) but incurs the highest computational overhead (105 ms latency and 145.7G FLOPs) by disrupting feature sparsity patterns. In contrast, our SDIMI achieves the most favorable trade-off, outperforming Voxel-DS by 0.2% in mIoU and 0.3% in NDS while maintaining superior efficiency (98 ms latency).

Furthermore, employing the BEV-level consistency loss $\mathcal{L}_{\text{bev}}$ enforces spatial alignment between segmentation and detection outputs, fostering more coherent multi-task predictions. Consequently, the performance metrics climb to 82.3% mIoU, 66.5% mAP, and 71.3% NDS. Since this auxiliary supervision is primarily utilized during training to refine the shared representation, it incurs negligible additional cost during inference.

Ultimately, we introduce the SIDMF module, which exploits instance-level priors to selectively propagate object-aware features from the detection branch to the segmentation head. This mechanism excels at recovering fine-grained object boundaries and refining semantic reasoning, as illustrated in Fig. 5. With the inclusion of SIDMF, our framework achieves state-of-the-art performance: 84.4% mIoU, 67.1% mAP, and 72.0% NDS. The final model operates with a latency of 107 ms and 156.1G FLOPs. The moderate increase in computational cost is a worthy trade-off for the significant gains in fine-grained segmentation accuracy. To further highlight the superiority of SIDMF, we benchmark it against alternative strategies in Table VI. For the naive baselines, where BEV features are simply projected back to the voxel space for addition or concatenation, the results are suboptimal. Concatenation even degrades Segmantation (82.3 mIou to 82.0 mIou) due to the introduction of background noise

TABLE VII
ORACLE ANALYSIS ON DEPENDENCY OF DETECTION QUALITY. WE COMPARE USING PREDICTED BOXES VS. GROUND TRUTH (GT) BOXES IN SIDMF.

Source of Bounding Boxes	mIoU	Δ
Predicted Boxes (Ours)	84.4	-
Ground Truth (Oracle)	**85.1**	+0.7

from the detection branch. Similarly, we adapted the Global Context Pooling (GCP) from LidarMultiNet [11], which utilizes global BEV context for feature enhancement. Although GCP improves general context, it still lags behind SIDMF by 0.9% mIoU. We attribute this to GCP's global aggregation mechanism, which risks blurring local fine-grained details. In contrast, SIDMF ensures precision by explicitly preserving instance-level structure.

G. Robustness of SIDMF

A primary concern in cascaded multi-task fusion is the risk of error coupling, where mislocalized detection boxes introduce noise into the segmentation branch. Given that SIDMF aggregates instance features via bounding box masks, we conduct two specific analyses to evaluate its robustness against detection imperfections.

Oracle Analysis: To assess the dependency on detection accuracy, we perform an "Oracle" experiment by replacing predicted bounding boxes with Ground Truth (GT) boxes during inference. As detailed in Table VII, using GT boxes yields an mIoU of 85.1%. The performance gap compared to our standard result of 84.4% is marginal. This indicates that our model is not overly reliant on perfect localization and that the current detection quality is sufficient to support high-performance segmentation.

Noise Sensitivity: We simulate detection jitter by injecting random spatial noise into the box centers. To ensure scale invariance across categories, we apply offsets proportional to object dimensions. Specifically, predicted center coordinates are perturbed by a ratio $\sigma \in \{2\%, 4\%, 6\%, 8\%\}$ relative to the object's dimension. As illustrated in Fig. 6, the model exhibits exceptional stability. Notably, with a 2% perturbation, the performance remains unchanged at 84.4% mIoU. Even under a significant perturbation ratio of 8%, the mIoU decreases by only 0.9%, demonstrating that the segmentation branch is resilient to localization errors.

This observed robustness is intrinsic to our design. First, the residual connection in (8) ensures that aggregated instance features merely refine—rather than replace—the original features F_{seg}. Second, the pooling operation within SIDMF smooths out local feature noise caused by box misalignment. Third, the end-to-end joint training paradigm inherently acts as a regularizer. Since the network encounters top-T imperfect proposals during training, segmentation gradients drive the MLP to learn noise suppression while concurrently optimizing the upstream detection branch for better localization. Together, these mechanisms effectively decouple segmentation performance from minor detection errors.

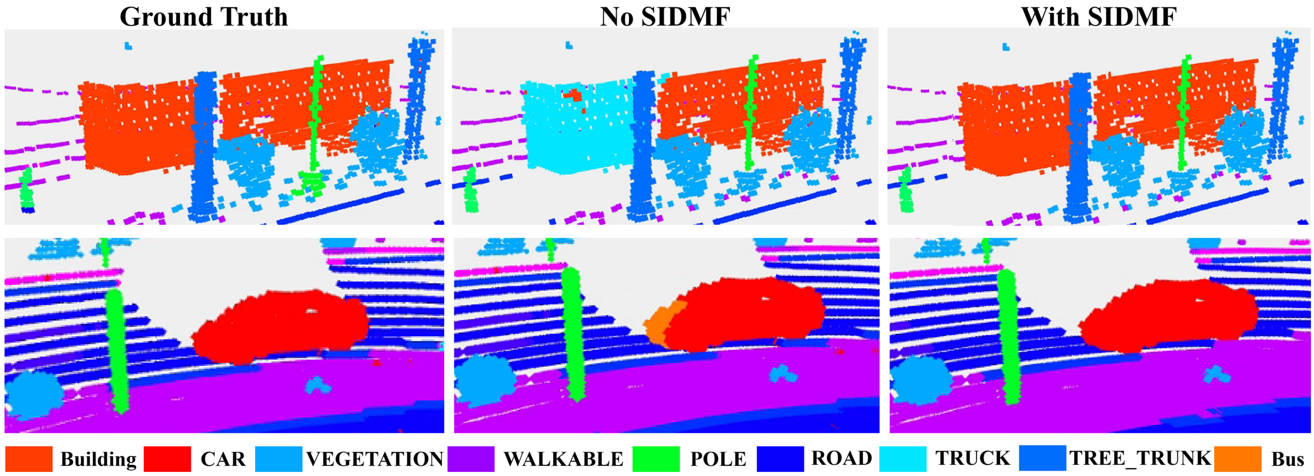

Fig. 5. Qualitative comparison of semantic segmentation results. The columns from left to right display the Ground Truth, the prediction without the SIDMF module (No SIDMF), and the prediction with the SIDMF module (With SIDMF). As observed, the baseline method (middle) suffers from inconsistent classification within objects. In contrast, our SIDMF module (right) effectively leverages instance-level priors to correct these errors, ensuring semantic consistency within object boundaries.

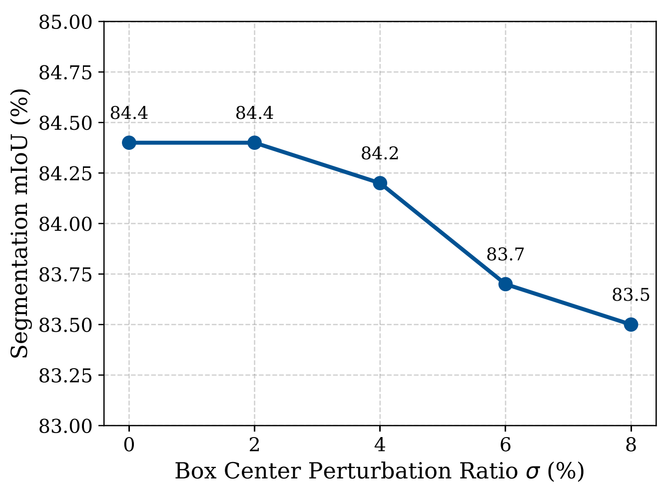

Fig. 6. Robustness analysis against box localization error. We inject random spatial offsets proportional to the object dimensions (2%, 4%, 6%, 8%) into the bounding box centers. The segmentation performance remains stable even under significant perturbations.

V. Conclusion

This paper addresses the challenges of restricted receptive fields and inadequate cross-task interaction in LiDAR multi-task perception by proposing an efficient framework that integrates receptive field expansion and cross-task feature fusion. The key innovations include the SDIMI and SIDMF modules. SDIMI employs density-agnostic strategies to hierarchically fuse multi-scale contextual features, expanding the receptive field while preserving feature sparsity. SIDMF leverages segmentation-to-detection bounding box masks to dynamically align instance-level features, enabling high-level feature propagation between tasks. Experimental results on NuScenes and Waymo Open Dataset validate the framework's superiority.

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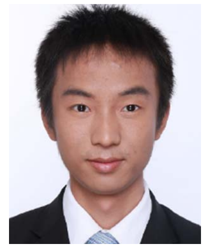

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